

LED Power Consumption

How the EHS Production Tool calculates LED wall power & current

This note explains, end to end, how the tool turns a panel's rated wattage into the power and amperage figures you see on the LED tab and in the project PDF. Every value below can be overridden per screen.

The build-up, step by step

1 Start with the panel's rated power

Every cabinet in the catalog has a rated wattage (its peak draw running full white). This is the starting point for one cabinet.

2 Scale it for brightness

You rarely run full brightness. The nominal calibration point is 5000 nits. Even at 0 nits a panel still pulls 30% of rated power (receiving cards, fans, processors never switch off) — that is the floor. Above the floor it scales linearly up to 100% at nominal.

```
brightnessFactor = 0.30 + 0.70 x (yourNits / 5000)    (capped at 1.0)
```

Example: a screen set to 2500 nits = $0.30 + 0.70 \times 0.5 = 65\%$ of rated power.

3 Add PSU overhead

Power supplies lose energy and you want a safety margin, so a percentage is added on top (default 25%, overridable per screen).

```
wattsPerCabinet = panel.power x brightnessFactor x (1 + overhead/100)
```

4 Multiply by enabled cabinets

Only cabinets that are actually powered count — disabled or blanked ones are excluded.

```
totalWatts = wattsPerCabinet x (number of enabled cabinets)
```

5 Convert watts to amps

Using the region's mains voltage (e.g. 230 V EU, 120 V US) and the power factor (default 0.95).

```
amps = totalWatts / (voltage x powerFactor)
```

6 Split across power chains

If you set a max-cabinets-per-chain, the tool works out how many chains you need and divides the amps evenly. A chain is flagged as overloaded when it exceeds 80% of the breaker rating (the standard de-rate).

Max output vs Average output

Max output = totalWatts above — peak draw, full white. Size your cable, distro and breakers off THIS number.

Average output = Max x 1/3. Real content (video, text, logos) almost never lights every pixel white, so realistic running draw is roughly a third of peak. Use it for generator fuel and heat planning — never for

sizing power infrastructure.

What you can override per screen

Voltage region - Power factor (default 0.95) - PSU overhead % (default 25%) - Brightness in nits - Cabinets per power chain. Each has a sensible default but can be set per screen.

In one line

Rated cabinet watts -> dim it for brightness -> pad it for PSU losses -> multiply by cabinet count -> divide by volts to get amps -> split across chains and check against breakers.